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United States Patent

12404326

Kind Code

B2

Date of Patent

September 02, 2025

Inventor(s)

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### **Bivalent bispecific antibody and preparation method thereof, coding gene, host cell and composition**

#### **Abstract**

Provided are a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. The bivalent bispecific antibody comprises: a) a single-chain variable fragment scFv, a flexible peptide, a heavy chain IgG1 constant region CH1 and a hinge region partial sequence of the antibody that specifically binds to a first antigen, and b) a single-chain variable fragment scFv, and a light chain constant region CL, that is, scFv1-CL or CL-scFv1, of the antibody that specifically binds to a second antigen; or comprises: c) a light chain and a heavy chain of the antibody that specifically binds to the first antigen, and d) a light chain and a heavy chain of the antibody that specifically binds to the second antigen.

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| <b>Appl. No.:</b>            | <b>17/418856</b>                                                                                                                                                                                           |
| <b>Filed (or PCT Filed):</b> | <b>December 26, 2019</b>                                                                                                                                                                                   |
| <b>PCT No.:</b>              | <b>PCT/CN2019/128582</b>                                                                                                                                                                                   |
| <b>PCT Pub. No.:</b>         | <b>WO2020/135555</b>                                                                                                                                                                                       |
| <b>PCT Pub. Date:</b>        | <b>July 02, 2020</b>                                                                                                                                                                                       |

## Prior Publication Data

### Document Identifier

US 20220073610 A1

### Publication Date

Mar. 10, 2022

## Foreign Application Priority Data

CN

201811622069.2

Dec. 28, 2018

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## Publication Classification

**Int. Cl.:** C07K16/28 (20060101)

**U.S. Cl.:**

CPC C07K16/2803 (20130101); C07K16/2827 (20130101); C07K2317/24 (20130101); C07K2317/31 (20130101); C07K2317/35 (20130101); C07K2317/622 (20130101); C07K2317/92 (20130101)

## Field of Classification Search

**USPC:** None

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## **Background/Summary**

### **CROSS REFERENCE TO RELATED APPLICATIONS**

(1) This application is a US National Phase application based upon PCT Application No. PCT/CN2019/128582, filed Dec. 26, 2019, which claims the priority of Chinese Patent Application No. 201811622069.2, filed to China National Intellectual Property Administration on Dec. 28, 2018, and titled with “BIVALENT BISPECIFIC ANTIBODY AND PREPARATION METHOD THEREOF, CODING GENE, HOST CELL AND COMPOSITION”, and the disclosures of which are hereby incorporated by reference.

**SEQUENCE LISTING**

(2) This application contains references to amino acid sequences and/or nucleic acid sequences which have been submitted concurrently herewith as the sequence listing text file entitled “69427904\_210033-APXU-GENSCI-Sequence\_listing-v,”, file size 56,643 bytes, created on Mar. 5, 2025. The aforementioned sequence listing is hereby incorporated by reference in its entirety.

#### FIELD

(3) The present disclosure relates to the technical field of antibody medicine, specifically to a bivalent bispecific antibody and preparation method thereof, coding gene, host cell and composition.

#### BACKGROUND

(4) A bispecific monoclonal antibody (BsAb) is a special antibody that is artificially made to bind two different antigens at the same time. Bispecific antibodies can recognize both tumor target cells and immune effector cells, so they have dual functions of antibody specificity and mediating the cytotoxicity of effector cells. Bispecific antibodies can recruit effector cells at tumor sites and activate effector cells to exert anti-tumor effects. The mechanism of killing tumor cells mediated by bispecific antibodies includes cell proliferation, cytokine release, cytotoxic peptides and regulation of enzymes. In vivo and clinical studies have proved that bispecific antibody-mediated immunotherapy can treat tumors in some animals, and clinically can mitigate the condition of patients with tumor and prolong their life. Therefore, the application of bispecific antibody-mediated immunocompetent cells in tumor therapy has a good prospect.

(5) Bispecific antibodies are not nature products and can only be prepared artificially. Bi- or multi-specific antibodies in the art can bind to at least two antigens and can be produced by cell fusion, chemical conjugation or recombinant DNA technology. Recently, a wide variety of recombinant bispecific antibody structures have been developed, such as tetravalent bispecific antibodies by fusion of, for example, an IgG antibody and a single-chain domain (Coloma, M. J., et al., *Nature Biotech.* 15 (1997) 159-163; WO2001077342; and Morrison, S. L., *Nature Biotech.* 25 (2007) 1233-1234). In addition, many other new forms that can bind to more than two antigens have been developed, in which the main structure of the antibody (IgA, IgD, IgE, IgG or IgM) is no longer limited to, such as diabodies, triabodies or tetrabodies, minibodies and several single-chain forms (scFv, Bis-scFv) (Holliger, P., et al., *Nature Biotech.* 23 (2005) 1126-1136; Fischer, N., and Léger, O., *Pathobiology* 74 (2007) 3-14; Shen, J., et al., *Journal of Immunological Methods* 318 (2007) 65-74; Wu, C., et al., *Nature Biotech.* 25 (2007) 1290-1297).

(6) In one method, the cell quadroma technology (Milstein, C. and A. C. Cuello, *Nature*, 305 (1983) 537-40) is utilized to produce a bispecific antibody that is very similar to a natural antibody. The cell quadroma technology is based on the somatic fusion of two different hybridoma cell lines expressing murine monoclonal antibodies with the desired bispecific antibody specificity. Because of the random pairing of two different heavy and light chains of antibodies in the hybridoma cell lines, up to 10 different antibody types will be generated, of which only one is the desired functional bispecific antibody. Due to the presence of mismatched by-products and significantly low yields, complicated purification procedures are required (Morrison, S. L., *Nature Biotech.* 25 (2007) 1233-1234). Generally, if recombinant expression technology is used, the same problem of mismatch by-products still exists.

(7) A method used to avoid the problem of mismatch by-products is called “knobs-into-holes”. The purpose is to force the heavy chains from two different antibodies to pair with each other by introducing mutation into the CH3 domain to modify the contact interface. In one chain, amino acids with large volume are replaced by amino acids with short side chains to form a “hole”. Conversely, amino acids with a large side chain are introduced to the other CH3 domain to form a “knob”. By co-expressing these two heavy chains, a higher yield of heterodimer form (“knob-hole”) compared with homodimer form (“hole-hole” or “knob-knob”) was observed (Ridgway, J. B., Presta, L. G., Carter, P. and WO 1996027011). The percentage of heterodimer can be further increased by reconstruction of the interaction interface of the two CH3 domains using phage

display method and introduction of disulfide bonds to stabilize the heterodimer (Merchant, A. M., et al., *Nature Biotech* 16 (1998) 677-681; Atwell, S., Ridgway, J. B., Wells, J. A., Carter, P., J. *Mol. Biol.* 270 (1997) 26-35). An important constraint of this strategy is that the light chains of the two parent antibodies must be the same to prevent mismatches and formation of inactive molecules.

(8) In addition to the “knob-hole” structure, the Fc pairing of different half-antibodies can also be achieved through the strand-exchange engineered domain (SEED) technology of IgG and IgA CH3 (Davis, J. H., et al., *Protein Eng. Des. Sel.*, 2010, 23 (4): 195-202).

(9) In order to solve the problem of the incorrect assembly of different light chains, a new process of double-cell line expressing half-antibodies separately and in vitro assembly has been developed recently. Inspired by the half-antibody random exchange process of human IgG4 antibodies naturally occurring under physiological conditions, GenMab has developed FAE (Fab-arm exchange) bifunctional antibody technology (Gramer, M. J., et al., *MAbs* 2013, 5 (6): 962-973.). Introducing two point mutations, K409R and F405L, into the IgG1 heavy chain CH3 domains of the two target antibodies respectively, can produce half-antibody exchange rearrangement similar to that of IgG4 antibodies. Two different IgG1 antibodies after mutation were expressed in two CHO cell lines respectively, and the assembly between the light and heavy chains of each half-antibody was completed. After protein A affinity purification, precise assembly between heterogeneous half-antibodies can be achieved in vitro by using a mild oxidant system.

(10) In addition to sharing light chains with the same sequence or performing in vitro assembly, the correct assembly of light chains of antibodies can also be facilitated by Crossmab technology. A representative product is Roche's Ang-2/VEGF CrossMab CH1-CL. Based on the modification of “knobs-into-holes”, Crossmab technology exchanged CL and CH1 in the Fab domain of Ang-2 antibody and remained the Fab domain of VEGF antibody unchanged. The light chain of the modified Ang-2 antibody is not easily mismatched with the heavy chain of the VEGF antibody, and the “knob-hole” structure can promote the heterodimerization of the two heavy chains (Schaefer, W, et al., *Proc Natl. Acad. Sci. USA*, 2011, 108 (27): 11187-11192).

(11) Moreover, two single-chain antibodies (scFv) or two Fabs can be linked through a peptide to form a bifunctional antibody fragment. A representative product is BiTE (bispecific T-cell engager) series products developed by Micromet in German. This series of products is generated by linking anti-CD3 single-chain antibodies with the single-chain antibodies against different anti-tumor cell surface antigens through a peptide (Baeuerle, P. A., et al., *Cancer Res.*, 2009, 69 (12): 4941-4944). The advantage of such antibody structure is that it has a small molecular weight, can be expressed in prokaryotic cells, and does not require the consideration of incorrect assembly; while the disadvantage is that it cannot mediate some corresponding biological functions due to a lack of antibody Fc fragment, and its half-life is short.

(12) The patent application publications US2015/0284475A1 and CN101896504A disclose bivalent bispecific antibodies, but the bivalent bispecific antibodies in the two applications have low affinity for antigens. The flexible peptide disclosed in “Prediction of three dimensional structure of bispecific antibodies linked by flexible peptide” published by Shao Changli, was used to modify the bivalent bispecific antibodies in the two aforementioned applications, but the affinity of the antibodies was still not ideal.

(13) In view of the problems of light chain mismatches, low correct assembly rate of light and heavy chains, and the too large or small molecule size of the above-mentioned bispecific antibodies, it is necessary to develop a new type of a bivalent bispecific antibody.

## SUMMARY

(14) In view of this, the present disclosure provides a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. The antibody has a less mismatched assembly rate of light and heavy chains and a moderate molecular size.

(15) In order to achieve the above objects of the present disclosure, the present disclosure provides the following technical solutions:

(16) The present disclosure provides a bivalent bispecific antibody, comprising a) a single-chain variable fragment (scFv) derived from an antibody that specifically binds to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, namely CH1-partial hinge-linker (flexible peptide)-scFv2 or scFv2-linker-CH1-partial hinge, and b) a single-chain variable fragment (scFv) derived from an antibody that specifically binds to a second antigen and a light chain constant region CL, namely scFv1-CL or CL-scFv1; or comprising c) a light chain and a heavy chain of an antibody that specifically binds to a first antigen, and d) a light chain and a heavy chain of an antibody that specifically binds to a second antigen, wherein light chain variable region of the light chain is linked to a flexible peptide and a linker, and heavy chain variable region of the heavy chain is linked to a heavy chain Fc fragment through a flexible peptide and a linker.

(17) Preferably, the flexible peptide comprises a sequence of (G4S/G4SAS)  $n$ , wherein  $n$  is an integer greater than or equal to 0, and the IgG1 partial hinge region linked to the flexible peptide comprises a sequence of EPKSCDK (SEQ ID NO: 24), wherein (G4S/G4SAS)  $n$  represents (GGGGS (SEQ ID NO: 25))  $n$ \_or (GGGGSAS (SEQ ID NO: 26))  $n$ , and  $n$  is an integer greater than or equal to 0.

(18) The linker comprises a sequence of L/GGGC (L/GGGC represents LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28)), and the first cysteine residue (C) in heavy chain hinge region linked to the linker is mutated to serine(S).

(19) Preferably, in steps a) and b), CL and CH1 form a heterodimer via a disulfide bond between the terminal cysteine residue in CL and the cysteine residue in the heavy chain hinge region.

(20) Preferably, in steps c) and d), the terminal cysteine residue in the linker LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28) linked to the heavy chain and the terminal cysteine residue in the linker LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28) linked to the light chain form a disulfide bond; and the CH3 domain of first heavy chain and the CH3 domain of second heavy chain are modified to a structure that facilitates the formation of the bivalent bispecific antibody.

(21) Preferably, the modification comprises a) modification to the CH3 domain of the first heavy chain: in the interface between the CH3 domain of the first heavy chain and the CH3 domain of the second heavy chain of the bivalent bispecific antibody, an amino acid residue in the CH3 domain of the first heavy chain is replaced with an amino acid residue with a volume larger than the original amino acid residue to form a knob in the CH3 domain of the first heavy chain, wherein the knob is capable of inserting into a hole of the CH3 domain of the second heavy chain, and b) modification to the CH3 domain of the second heavy chain: in the interface between the CH3 domain of the second heavy chain and the CH3 domain of the first heavy chain of the bivalent bispecific antibody, an amino acid residue in the CH3 domain of the second heavy chain is replaced with an amino acid residue with a volume smaller than the original amino acid residue to form a hole in the CH3 domain of the second heavy chain, wherein the hole is capable of holding the knob of the CH3 domain of the first heavy chain; wherein each of the first heavy chain variable region and the light chain variable region is linked to (GGGGS (SEQ ID NO: 25))  $n$ \_or (GGGGSAS (SEQ ID NO: 26))  $n$  and LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28), and a disulfide bond is formed between the cysteine residues of the two L/GGGC.

(22) Preferably, the amino acid residue with a volume larger than the original amino acid residue is selected from the group consisting of arginine (R), phenylalanine (P), tyrosine (Y), and tryptophan (W).

(23) The amino acid residue with a volume smaller than the original amino acid residue is selected from the group consisting of alanine (A), serine(S), threonine (T), and valine (V).

(24) Preferably, the bivalent bispecific antibody provided by the present disclosure comprises: a) a single-chain variable fragment (scFv), a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, derived from an antibody that specifically binds to a first antigen, i.e., the amino acid sequence as shown in SEQ ID NO: 2, and b) a single-chain variable fragment (scFv)

and a light chain constant region CL derived from an antibody that specifically binds to a second antigen, i.e., the amino acid sequence as shown in SEQ ID NO: 11; or comprises c) a light chain and a heavy chain of an antibody that specifically binds to a first antigen, i.e., the amino acid sequences as shown in SEQ ID NO: 5 and SEQ ID NO: 14, and d) a light chain and a heavy chain of an antibody that specifically binds to a second antigen, i.e., the amino acid sequences as shown in SEQ ID NO: 6 and SEQ ID NO: 15, wherein the light chain variable region is linked to a flexible peptide and a linker, and the heavy chain variable region is linked to the heavy chain Fc fragment through a flexible peptide and a linker.

(25) The present disclosure also provides a method for producing the bivalent bispecific antibody, comprising the following steps: a) Transforming a host cell with the following vectors: a first vector (comprising the gene encoding SEQ ID NO: 2), comprising a nucleic acid encoding a single-chain variable fragment and a heavy chain constant region CH1 of an antibody that specifically binds to a first antigen, and a second vector (comprising the gene encoding SEQ ID NO: 11), comprising a nucleic acid encoding a single-chain variable fragment and a light chain constant region of an antibody that specifically binds to a second antigen; or with the following vectors: a third vector (comprising the gene encoding SEQ ID NO: 14), comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a first antigen, a fourth vector (comprising the gene encoding SEQ ID NO: 5), comprising a nucleic acid encoding a heavy chain of an antibody that specifically binds to a first antigen, a fifth vector (comprising the gene encoding SEQ ID NO: 15), comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a second antigen, wherein the light chain variable region is linked to a linker, and a sixth vector (comprising the gene encoding SEQ ID NO: 6), comprising a nucleic acid encoding a heavy chain of an antibody that specifically binds to a second antigen, wherein the heavy chain variable region is linked to the heavy chain Fc fragment through a linker; b) culturing the host cell under conditions that allow the synthesis of an antibody; and c) recovering the antibody from the culture.

(26) The present disclosure also provides a nucleic acid molecule encoding the bivalent bispecific antibody of the present disclosure, comprising a first nucleic acid (the gene encoding SEQ ID NO: 2), encoding a single-chain variable fragment (scfv), a flexible peptide, a heavy chain igg1 constant region CH1, and a partial hinge region of an antibody that specifically binds to a first antigen, and a second nucleic acid (the gene encoding SEQ ID NO: 11), encoding a single-chain variable fragment (scfv) and a light chain constant region CL of an antibody that specifically binds to a second antigen; or comprising a third nucleic acid (the gene encoding SEQ ID NO: 14), encoding a light chain of an antibody that specifically binds to a first antigen, a fourth nucleic acid (the gene encoding SEQ ID NO: 5), encoding a heavy chain of an antibody that specifically binds to a first antigen, a fifth nucleic acid (the gene encoding SEQ ID NO: 15), encoding a light chain of an antibody that specifically binds to a second antigen, wherein the light chain variable region is linked to a flexible peptide and a linker, and a sixth nucleic acid (the gene encoding SEQ ID NO: 6), encoding a heavy chain of an antibody that specifically binds to a second antigen, wherein the heavy chain variable region is linked to the heavy chain Fc fragment through a flexible peptide and a linker.

(27) The present disclosure also provides a host cell, comprising a first vector, comprising a nucleic acid encoding a single-chain variable fragment and a heavy chain constant region CH1 of an antibody that specifically binds to a first antigen, and a second vector, comprising a nucleic acid encoding a single-chain variable fragment and a light chain constant region of an antibody that specifically binds to a second antigen; or comprising a third vector, comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a first antigen, a fourth vector, comprising a nucleic acid encoding a heavy chain of an antibody that specifically binds to a first antigen, a fifth vector, comprising a nucleic acid encoding a light chain of an antibody that specifically binds to a second antigen, wherein the light chain variable region is linked to a linker, and a sixth vector, comprising a nucleic acid encoding a heavy chain of an antibody that



specifically binds to a second antigen, wherein the heavy chain variable region is linked to the heavy chain Fc fragment through a linker.

(28) The present disclosure also provides a composition comprising the bivalent bispecific antibody, and the composition is a pharmaceutical composition or a diagnostic composition.

(29) Preferably, the pharmaceutical composition also comprises at least one pharmaceutical excipient.

(30) The present disclosure provides a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. The bivalent bispecific antibody comprises a) a single-chain variable fragment (scFv), a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, derived from an antibody that specifically binds to a first antigen, namely CH1-partial hinge-linker (flexible peptide)-scFv2 or scFv2-linker-CH1-partial hinge, and b) a single-chain variable fragment (scFv) and a light chain constant region CL derived from an antibody that specifically binds to a second antigen, namely scFv1-CL or CL-scFv1; or comprises c) a light chain and a heavy chain of an antibody that specifically binds to a first antigen, and d) a light chain and a heavy chain of an antibody that specifically binds to a second antigen, wherein light chain variable region of the light chain is linked to a flexible peptide and a linker, and heavy chain variable region of the heavy chain is linked to the heavy chain Fc fragment through a flexible peptide and a linker.

(31) The present disclosure has the following technical effects.

(32) The bivalent bispecific antibody of the present disclosure has high affinities to the first antigen molecule and the second antigen molecule, which are equivalent to the parent monoclonal antibody molecule. For example, the affinities of B2 and FV1 to PD-L1 are  $1.09\text{E-}10$  M and  $2.71\text{E-}10$  M respectively; and the affinities of B2 and FV1 to PD-L1 are  $2.58\text{E-}8$  M and  $1.50\text{E-}8$  M, respectively. It can be seen that the antibody of the present disclosure has a high correct assembly rate of light and heavy chains and a moderate molecular size.

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## Description

### BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a schematic diagram of bivalent bispecific antibodies with B structure, including four structures: B1, B2, B3 and B4.

(2) FIG. 2 is a schematic diagram of bivalent bispecific antibodies with FV structure, including four structures: FV1, FV2, FV3 and FV4. In the structure FV1, there is no flexible peptide; in the structure FV2, the sequence of the flexible peptide is GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26); in the structure FV3, the sequence of the flexible peptide is GGGSGGGSGGGGS (SEQ ID NO: 29) or GGGGSASGGGGSASGGGGSAS (SEQ ID NO: 30); and in the structure FV4, the sequence of the flexible peptide is GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26).

(3) FIG. 3 shows the SDS-PAGE results of the transient expression of bispecific antibodies with B structure and with FV structure. M indicates Marker; Lanes 1, 3, 5, 7, 9, 11, 13, and 15 represent the non-reducing electrophoresis results of B1, B2, B3, B4, FV1, FV2, FV3, and FV4, respectively; and Lanes 2, 4, 6, 8, 10, 12, 14, and 16 represent the reducing electrophoresis results of B1, B2, B3, B4, FV1, FV2, FV3 and FV4, respectively.

(4) FIG. 4 shows the ELISA results of bivalent bispecific antibodies with B structure and with FV structure, including the ELISA results of four B structures of B1, B2, B3 and B4 (FIG. 4A) and four FV structures of FV1, FV2, FV3 and FV4 (FIG. 4B).

(5) FIG. 5 shows the SDS-PAGE results of the purified bispecific antibodies with preferred B2 structure and with preferred FV1 structure. M indicates Marker; Lanes 1, 2, and 3 represent the supernatants from the non-reducing transient expression of B2 bispecific antibody, the non-

reducing eluent of B2 bispecific antibody by Protein L affinity chromatography, and the non-reducing eluent of B2 bispecific antibody by PD-L1 affinity chromatography; Lanes 4, 6, 8 represent the non-reducing eluent of FV1 bispecific antibody by Mab SelectSure affinity chromatography, the non-reducing eluent of FV1 bispecific antibody by Protein L affinity chromatography, and the non-reducing eluent of FV1 bispecific antibody by hCD47 affinity chromatography; and Lanes 5, 7, and 9 represent the reducing eluent of FV1 bispecific antibody by Mab SelectSure affinity chromatography, the reducing eluent of FV1 bispecific antibody by Protein L affinity chromatography, and the reducing eluent of FV1 bispecific antibody by hCD47 affinity chromatography.

(6) FIG. 6 shows the ELISA results of the bivalent bispecific antibodies with preferred B2 structure and with FV1 structure after purification.

#### DETAILED DESCRIPTION

(7) The present disclosure discloses a bivalent bispecific antibody and a preparation method thereof, a coding gene, a host cell and a composition. In view of the content herein, those skilled in the art can make appropriate modifications to the process parameters. It should be particularly indicated that, all similar replacements and changes are obvious for those skilled in the art, which are deemed to be included in the present disclosure. The methods and uses of the present disclosure have been described by way of preferred embodiments, and it will be apparent to those skilled in the art that changes as well as appropriate modifications and combinations of the methods and uses described herein may be made without departing from the content, spirit and scope of the present disclosure, to achieve and apply the techniques of the present disclosure.

(8) The terms used herein are explained as follows.

(9) The term “antibody” used herein refers to an intact, monoclonal antibody. The antibody comprises two pairs of “light chain” (LC) and “heavy chain” (HC). The light chain and heavy chain of the antibody are polypeptides composed of several domains. In an intact antibody, each heavy chain comprises a heavy chain variable region VH and a heavy chain constant region. The heavy chain constant region includes the heavy chain constant domains CH1, CH2, and CH3 (antibody types IgA, IgD, and IgG) and optionally the heavy chain constant domain CH4 (antibody types IgE and IgM). Each light chain comprises a light chain variable region VL and a light chain constant region CL. An example of naturally occurring intact antibodies are the IgG antibodies. The variable regions VH and VL can be further divided into hypervariable regions, called complementarity determining regions (CDRs), and there are regions distributed between CDR which are more conservative, called framework regions (FRs). Each VH and VL consists of three CDRs and four FRs, arranged from the amino terminal to the carboxy terminal in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4 (Janeway, C. A., Jr., et al., (2001) Immunobiology, 5th edition, Garland Press; and Woof, J., Burton D. Nat. Rev. Immunol. 4 (2004) 89-99). The two pairs of heavy and light chains can specifically bind to the same antigen. Therefore, the intact antibody is a bivalent, monospecific antibody. The “antibody” includes, for example, mouse antibodies, human antibodies, chimeric antibodies, humanized antibodies, and genetically engineered antibodies, provided that their unique properties are maintained. The particularly preferred antibodies are human or humanized antibodies, especially recombinant human or humanized antibodies.

(10) There are five types of mammalian antibody heavy chain, represented by Greek letters:  $\alpha$ ,  $\delta$ ,  $\epsilon$ ,  $\gamma$ , and  $\mu$  (Janeway, C. A., Jr. et al., (2001) Immunobiology, 5th edition, Garland Press). The type of heavy chain corresponds to the type of antibody, and these chains are present in IgA, IgD, IgE, IgG, and IgM antibodies, respectively (Rhoades, R. A., Pflanzner, R.G. (2002), Human Physiology, 4th edition, Thommesen knowledge). Different heavy chains differ in size and composition. The heavy chains  $\alpha$  and  $\gamma$  contain about 450 amino acids, while  $\mu$  and  $\delta$  contain about 550 amino acids.

(11) Each heavy chain has two types of region, the constant region and the variable region. The constant region is the same in all antibodies of the same isotype, but different in antibodies of different isotypes. The heavy chains  $\gamma$ ,  $\alpha$  and  $\delta$  have a constant region composed of three constant

domains CH1, CH2 and CH3 and a hinge region for increasing flexibility (Woof, J., Burton D. Nat. Rev. Immunol. 4 (2004) 89-99). The heavy chains  $\mu$  and  $\delta$  have a constant region composed of 4 constant domains CH1, CH2, CH3 and CH4 (Janeway, C. A., Jr. et al., (2001) Immunobiology, 5th edition, published by Garland Society). The variable region of the heavy chain differs in antibodies produced by different B cells, but is the same for all antibodies produced by a single B cell or B cell clone. The variable region of each heavy chain is approximately 110 amino acids in length and consists of a single antibody domain.

(12) In mammals, there are only two types of light chains, which are called  $\lambda$  and  $\kappa$ . The light chain comprises two continuous domains: one constant domain CL and one variable domain VL. The light chain is approximate 211-217 amino acids in length. Preferably, the light chain is a  $\kappa$  light chain, and the constant domain CL is preferably CK.

(13) The term “monoclonal antibody” or “monoclonal antibody composition” used herein refers to an antibody preparation of a single amino acid composition.

(14) The “antibody” according to the present disclosure can be of any type (such as IgA, IgD, IgE, IgG and IgM, preferably IgG or IgE), or subtype (such as IgG1, IgG2, IgG3, IgG4, IgA1 and IgA2, preferably IgG1). The bivalent bispecific antibody according to the present disclosure is derived from two antibodies with the Fc fragments which are of the same subtype (such as IgG1, IgG4, etc., preferably IgG1), preferably Fc fragments of the same allotype.

(15) The “Fc fragment of an antibody” or “Fc fragment” is a term well known to those skilled in the art and is defined based on the papain cleavage of the antibody. The antibody according to the present disclosure comprises, for example, Fc fragment, preferably the Fc fragment derived from a human antibody and preferably all other parts of a human constant region. The Fc fragment of the antibody is directly involved in complement activation, Clq binding, C3 activation and Fc receptor binding. Although the effect of the antibody on the complement system depends on certain conditions, the binding to Clq is caused by the binding sites in the Fc fragment. The binding sites are prior art and described in, for example, Lukas, T. J., et al., J. Immunol. 127 (1981) 2555-2560; Brunhouse, Cebra, J. J., Mo 1. Immunol. 16 (1979) 907-917; Burton, D. R., et al., Nature 288 (1980) 338-344; Thommesen, J. E., et al., Mol. Immunol. 37 (2000) 995-1004; Idusogie, E. E., et al., J. Immunol. 164 (2000) 4178-4184; Hezareh, M., et al., J. Virol. 75 (2001) 12161-12168; Morgan, A., et al., Immunology 86 (1995) 319-324; and EP0307434. The binding sites are, for example, L234, L235, D270, N297, E318, K320, K322, P331 and P329 (according to Kabat's EU catalog number). Antibodies of subtypes IgG1, IgG2, and IgG3 usually exhibit complement activation, Clq binding and C3 activation, while IgG4 neither activates the complement system, nor binds to Clq or activates C3. Preferably, the Fc fragment is a human Fc fragment.

(16) As used herein, the term “recombinant human antibody” includes all human antibodies prepared, expressed, produced or isolated by recombinant methods, such as antibodies isolated from host cells, such as from NSO or CHO cells, or isolated from human immunoglobulin genes of transgenic animal, or antibodies expressed by recombinant expression vectors transfected into host cells. This recombinant human antibody has constant and variable regions in a rearranged form.

(17) As used herein, a “variable domain” refers to each pair of light chain and heavy chain that directly participates in the binding of an antibody to an antigen. The domains of human variable light chain and heavy chain have the same general structure and each domain comprises four framework regions (FR), which have generally conserved sequence and connected by three hypervariable regions (CDRs). The framework region adopts a self-folding conformation and the CDR can form a loop connecting the self-folding structure. The CDRs in each chain are maintained in their three-dimensional structure by the framework regions and form an antigen binding site together with the CDRs from the other chain. The CDR3 regions of heavy chain and light chain of antibody play a particularly important role in the binding specificity/affinity of the antibodies of the present disclosure.

(18) As used herein, the term “hypervariable region” or “antigen-binding fragment of an antibody”

refers to amino acid residues of the antibody that are responsible for antigen binding. The hypervariable region comprises amino acid residues from the “complementarity determining region”. “Framework regions” are those regions of variable domains other than the hypervariable region residues defined herein. Therefore, the light chain and heavy chain of the antibody from N-terminal to C-terminal include the domains FR1, CDR1, FR2, CDR2, FR3, CDR3, and FR4, and the CDRs on each chain are separated by the framework amino acids. In particular, CDR3 of the heavy chain is the most conducive region for antigen binding. The CDR and FR regions are determined according to the standard definition of Kabat et al., Sequences of Proteins of Immunological Interest, 5th edition, Public Health Service, National Institutes of Health, Bethesda, MD, (1991).

(19) The “constant regions” of the heavy and light chains do not directly participate in the binding of antibodies to antigens but exhibit multiple effector functions.

(20) As used herein, the term “bivalent bispecific antibody” refers to an antibody as described above, in which two single-chain variable fragments specifically bind to different antigens, that is, the first single-chain variable fragment and the light chain constant region specifically bind to the first antigen, while the second single-chain variable fragment and the heavy chain constant region 1 and partial hinge region specifically bind to the second antigen. The bivalent bispecific antibody can specifically bind to two different antigens but no more than two antigens at the same time. In contrast, on one hand, a monospecific antibody that can only bind to one antigen, and on the other hand, a tetravalent, tetraspecific antibody that can bind four antigen at the same time.

(21) As used herein, the terms “antigen” and “antigen molecule” are used interchangeably and refer to all molecules that can be specifically bound by an antibody. The bivalent bispecific antibody specifically binds to a first antigen and a second different antigen. As used herein, the term “antigen” includes, for example, proteins, different epitopes on the protein (as different antigens within the meaning of the present disclosure), and polysaccharides, which mainly includes bacteria, viruses and parts of other microorganisms (shell, envelope, cell wall, flagella, fimbriae and toxins). Lipids and nucleic acids are antigenic only when combined with proteins and polysaccharides. Non-microbial exogenous (non-self) antigens may include pollen, egg white, and proteins from transplanted tissues and organs or proteins on the surface of infused blood cells. Preferably, the antigen is selected from the group consisting of cytokines, cell surface proteins, enzymes and receptor.

(22) Tumor antigens are those antigens contained in MHC I or MHC II molecules on the surface of tumor cells. These antigens can sometimes be presented by tumor cells, and never by normal cells. As such, they are called tumor-specific antigens (TSAs) and are typically produced by tumor-specific mutations. More common are antigens presented by tumor cells and normal cells, and they are called tumor-associated antigens (TAAs). Cytotoxic T lymphocytes that recognize these antigens may be able to destroy tumor cells before they proliferate or metastasize. Tumor antigens can also be present on the tumor surface in the form of, for example, mutant receptors, in which case they should be recognized by B cells.

(23) In a preferred embodiment, at least one of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds is a tumor antigen.

(24) In another preferred embodiment, both of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds are tumor antigens. In this case, the first and second antigens may also be the two different epitopes on the same tumor-specific protein.

(25) In another preferred embodiment, one of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds is a tumor antigen, and the other is an effector cell antigen, such as a T cell receptor, CD3, CD16, etc.

(26) In another preferred embodiment, one of the two different antigens (the first and second antigens) to which the bivalent bispecific antibody specifically binds is a tumor antigen, and the

other is an anti-cancer substance such as a toxin or a kinase inhibitor.

(27) As used herein, “specific binding” or “specifically bind” refers to an antibody that specifically binds to an antigen. Preferably, the binding affinity of the antibody that specifically binds to the antigen has a  $K_D$  value below  $10^{-9}$  mol/L, such as  $10^{-10}$  mol/L, preferably a  $K_D$  value below  $10^{-10}$  mol/L, such as  $10^{-12}$  mol/L. Binding affinity is determined using standard binding assays, such as surface plasmon resonance technology (Biacore).

(28) The term “epitope” comprises any polypeptide determinant capable of specifically binding an antibody. In certain embodiments, epitope determinants include chemically polar surface groupings of molecules, such as amino acids, sugar side chains, phosphoryl or sulfonyl groups. In certain embodiments, an epitope can have specific three-dimensional structural characteristics and/or specific charging characteristics. An epitope is a region of an antigen bound by an antibody. In certain embodiments, when an antibody preferably recognizes its target antigen in a complex mixture of proteins and/or macromolecules, the antibody is said to specifically bind to the antigen.

(29) As used herein, the term “nucleic acid” or “nucleic acid molecule” includes DNA molecules and RNA molecules. The nucleic acid molecule may be single-stranded or double-stranded, but is preferably double-stranded DNA.

(30) As used herein, the expressions “cell”, “cell line” and “cell culture” are used interchangeably, and all of these include progeny. Therefore, the terms “transformant” and “transformed cells” include primary subject cells and cultures derived therefrom, regardless of the number of generation. It is also understood that the DNA content of all offspring may not be exactly the same due to intentional or unintentional mutations. They also include the mutant offspring with the same function or biological activity screened in the originally transformed cells. When different names are meant, it will be clear due to the context.

(31) As used herein, the term “transformation” refers to the process of transferring a vector/nucleic acid into a host cell. If cells without an insurmountable cell wall barrier are used as host cells, transfection is performed, for example, by the calcium phosphate precipitation method as described in Graham and van der Eh, *Virology* 52 (1978) 546ff. However, other methods of introducing DNA into cells can also be used, such as by nuclear injection or by protoplast fusion. If prokaryotic cells or cells containing substantial cell wall structures are used, transfection is performed, for example, by calcium treatment with calcium oxide, as described in Cohen, F. N., et al., *PNAS*. 69 (1972) 7110ff.

(32) The use of transformation recombination to generate antibodies is well known in the prior art and is described in, for example, the review article Makrides, S. C., *Protein Expr. Purif.* 17 (1999) 183-202; Geisse, S., et al., *Protein Expr. Purif.* 8 (1996) 271-282; Kaufman, R. J., *Mol. Biotechnol.* 16 (2000) 151-161; Werner, R. G., et al., *Arzneimittel Forschung* 48 (1998) 870-880 and U.S. Pat. Nos. 6,331,415 and 4,816,567.

(33) As used herein, “expression” refers to the process of transcribing nucleic acid into mRNA and/or the process of subsequently translating the transcribed mRNA into peptides, polypeptides or proteins. The transcript and the encoded polypeptide are collectively called the gene product. If the polynucleotide is derived from genomic DNA, expression in eukaryotic cells may include splicing of mRNA.

(34) A “vector” is a nucleic acid molecule, especially self-replicating one, which transfers the inserted nucleic acid molecule into and/or between host cells. The term includes vectors whose primary function is to insert DNA or RNA into cells, replication vectors whose primary function is to replicate DNA or RNA, and expression vectors whose function is to transcribe and/or translate DNA or RNA. It also includes vectors that provide more than one of the above-mentioned functions.

(35) An “expression vector” is a polynucleotide, which can be transcribed and translated into a polypeptide after being introduced into a suitable host cell. “Expression system” generally refers to a suitable host cell including an expression vector whose function is to produce a desired

expression product.

(36) The bivalent bispecific antibody according to the present disclosure is preferably produced by a recombinant method. The method is generally known in the art and includes protein expression in prokaryotic and eukaryotic cells, subsequent isolation of antibody polypeptides and generally purification to pharmaceutical purity. For protein expression, nucleic acids encoding light and heavy chains or fragments thereof are inserted in an expression vector by standard methods. The expression is carried out in suitable prokaryotic or eukaryotic host cells such as CHO cells, NSO cells, SP2/0 cells, HEK293 cells, COS cells, yeast or *E. coli* cells, and the antibody is recovered from the cells (the supernatant or cells after lysis). The bivalent bispecific antibody may exist in whole cells, as cell lysates, or in partially purified or substantially pure form. Purification is carried out by standard techniques, including alkali/SDS treatment, column chromatography and other techniques well known in the art to eliminate other cellular components or other contaminants, such as other cellular nucleic acids or proteins. Reference could be made to Ausubel, F., et al., *Current Protocols in Molecular Biology*, Greene Publishing and WileyInterscience, New York (1987). The expression in NSO cells is described in, for example, Barnes, L. M., et al., *Cytotechnology* 32 (2000) 109-123; and Barnes, L. M., et al., *Biotech. Bioeng.* 73 (2001) 261-270. Transient expression is described in, for example, Durocher, Y., et al., *Nucl. Acids Res.* 30 (2002) IV. The cloning of variable domains is described in Orlando, R., et al., *Proc. Natl. Acad. Sci. USA* 86 (1989) 3833-3837; Carter, P., et al., *Proc. Natl. Acad. Sci. USA* 89 (1992) 4285-4289; and Norderha, L., et al., *J. Immunol. Methods* 204 (1997) 77-87. A preferred transient expression system (HEK 293) is described in Schlaeger, E. J., and Christensen, K., *Cytotechnology* 30 (1999) 71-83 and Schlaeger, E. J., *J. Immunol. Methods* 194 (1996) 191-199.

(37) Control sequences suitable for prokaryotes comprise, for example, promoters, optional operon sequences, and ribosome binding sites. It is known that eukaryotic cells utilize promoters, enhancers and polyadenylation signals.

(38) A nucleic acid is "operably linked" when placed in a functional relationship with another nucleic acid sequence. For example, the DNA of the polypeptide is operably linked to a leader sequence or a DNA secreting leader sequence, provided that it is expressed as a preprotein involved in the secretion of the polypeptide; the coding sequence is operably linked to a promoter or enhancer, provided that it affects the transcription of the sequence; or the coding sequence is operably linked to a ribosome binding site, provided that it is positioned to facilitate translation. Generally, "operably linked" means that the DNA sequence being linked is continuous, and in the case of being linked to sequence secreting a leader, is continuous and in an open reading frame. However, the enhancer does not have to be continuous. Linkage is achieved by linking at convenient restriction sites. If the site is not present, the synthetic oligonucleotide adaptor or linker is used according to conventional practice.

(39) By conventional immunoglobulin purification procedures, such as, for example, protein A-agarose, hydroxyapatite chromatography, gel electrophoresis, dialysis, or affinity chromatography, the bivalent bispecific antibody will be appropriately isolated from the culture medium. The DNA and RNA encoding monoclonal antibodies can be easily isolated and sequenced using conventional procedures. Quadroma cells can function as the source of the DNA and RNA. Once isolated, the DNA can be inserted into an expression vector, which is then transfected into host cells that would otherwise not produce immunoglobulin, such as HEK293 cells, CHO cells, or myeloma cells, to generate the recombinant monoclonal antibodies in the host cells.

(40) Amino acid sequence variants (or mutants) of bivalent bispecific antibodies are prepared by introducing appropriate nucleotide changes into the antibody DNA, or by nucleotide synthesis. However, such modification can only be performed in a very limited range, such as the range described above. In addition, the modification does not change the above-mentioned antibody characteristics, such as IgG isotype and antigen binding, but can improve the yield of recombinant products or protein stability or promote purification.

(41) Raw materials or reagents, used in the bivalent bispecific antibody and the preparation method thereof, the coding gene, host cell and composition provided by the present disclosure, can all be purchased from the market.

(42) The present disclosure will be further illustrated by the following examples:

#### Example 1 Preparation of Bivalent Bispecific Antibodies

(43) 1. Construction of transient transfection expression vectors for bivalent bispecific antibodies

(44) (1) Materials

(45) The sequence of VL (SEQ ID NO: 18) and the sequence of VH (SEQ ID NO: 19) are derived from the anti-human CD47 humanized monoclonal antibody 059-4.16.2 H1L2, which was obtained by humanization of murine antibody (CN 201610436519.3) obtained from quadromas.

(46) The sequence of VL (SEQ ID NO: 20) and the sequence of VH (SEQ ID NO: 21) was derived from the anti-human PD-L1 humanized monoclonal antibody 047 Ab-6, which was obtained by panning the natural human antibodies library (CN 201810044303.1). The coding nucleotides of heavy chain constant region CH1, hinge region and Fc of IgG1, and nucleotides of Kappa chain constant region are derived from human IgG1.

(47) (2) Methods

(48) pGS003 was selected to construct the expression vectors for the heavy chain and light chain of bivalent bispecific antibodies (8 antibodies, the schematic diagrams of the structure are shown in FIG. 1 and FIG. 2). Primers were designed according to the coding nucleotides of VL and VH derived from the anti-human CD47 humanized monoclonal antibody 059-4.16.2 H1L2, the coding nucleotides of VL and VH derived from the anti-human PD-L1 humanized monoclonal antibody 047 Ab-6, the coding nucleotides of heavy chain constant region CH1, hinge region and Fc of IgG1, and nucleotide sequence of Kappa chain constant region, and multiple cloning sites in the vector. After PCR amplification, 9 heavy chain coding sequences and 8 light chain coding sequences were respectively cloned into pGS003 by in vitro recombination method (Synbio Technologies, iMulli Multi-fragment Recombination Cloning Kit), as shown in Table 1. After sequencing to identify the correct insertion of the antibody gene, the recombinant expression vector was transformed into *E. coli* TOP10F'. Then single colonies was picked and inoculated in LB medium containing 100 µg/mL of ampicillin, and cultured with shaking at 37° C. for 16 hours. The plasmids were extracted using endotoxin-removal, large-scale extraction kit of Zymo Research, and the obtained plasmids were dissolved in 1 mL of ultrapure water, and the plasmid concentration and OD260/280 were determined with a spectrophotometer. A plasmid with OD260/280 value between 1.8 and 1.9 is relatively pure plasmid DNA.

(49) TABLE-US-00001 TABLE 1 List of transient transfection expression vectors for heavy and light chains

| Heavy chain  | Heavy chain   | Light chain | Light chain   | expression amino acid | expression amino acid |
|--------------|---------------|-------------|---------------|-----------------------|-----------------------|
| vector       | sequence      | vector      | sequence      | H1                    | L1                    |
| SEQ ID NO: 1 | SEQ ID NO: 10 | H2          | SEQ ID NO: 2  | L2                    | SEQ ID NO: 11         |
| H3           | SEQ ID NO: 3  | L3          | SEQ ID NO: 12 | H4                    | SEQ ID NO: 4          |
| L4           | SEQ ID NO: 13 | H5          | SEQ ID NO: 5  | L5                    | SEQ ID NO: 14         |
| H6           | SEQ ID NO: 6  | L6          | SEQ ID NO: 15 | H7                    | SEQ ID NO: 7          |
| L7           | SEQ ID NO: 16 | H8          | SEQ ID NO: 8  | L8                    | SEQ ID NO: 17         |
| H9           | SEQ ID NO: 9  |             |               |                       |                       |

2. Transfection, Expression and Detection in Mammalian 293E Cells

(50) In the above 9 heavy chain expression vectors and 8 light chain expression vectors, H1, L2, L3 and H4 were used to express scFv of anti-hCD47, L1, H2, H3 and L4 were used to express scFv of anti-hPD-L1, H5 and L5 were used to express VH and VL of anti-hPD-L1, respectively, H6, H7, H8 and H9 were used to express VH of anti-hCD47, and L6, L7 and L8 were used to express VL of anti-hCD47.

(51) After combining the above vectors in accordance with H1+L1 (B1 structure), H2+L2 (B2 structure), H3+L3 (B3 structure), H4+L4 (B4 structure), H5+L5+H6+L6 (FV1 structure), H5+L5+H7+L7 (FV2 structure), H5+L5+H8+L8 (FV3 structure) and H5+L5+H9+L7 (FV4 structure), 2 mL 293E system of transient transfection expression was performed for evaluation. As to flexible peptide and linker, FV1 was L/GGGC (LGGC (SEQ ID NO: 27) or GGGC (SEQ ID

NO: 28)), FV2 was GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26)+LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28), FV3 was GGGSGGGSGGGGS (SEQ ID NO: 29) or GGGGSASGGGSASGGGSAS (SEQ ID NO: 30)+LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28), and for FV4, the light chain was GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26)+LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28) and the heavy chain was LGGC (SEQ ID NO: 27) or GGGC (SEQ ID NO: 28)+GGGGS (SEQ ID NO: 25) or GGGGSAS (SEQ ID NO: 26). The expression level was detected and the binding between the antibody and human CD47 or human PD-L1 was detected by ELISA. The results are shown in FIG. 3 and FIG. 4 (wherein 4A represents the ELISA detection result of the bivalent bispecific antibody with B structure, and 4B represents the ELISA detection result of the bivalent bispecific antibody with FV structure). B2 structure performs the best in the expression, assembly and antigen binding among the bispecific antibodies with B-type structure; and FV1 structure performs the best in the expression, assembly and antigen binding among the bispecific antibodies with FV-type structure. Therefore, B2 and FV1 are preferred antibodies.

(52) 293E cells were used to perform amplified transient transfection expression of B2 and FV1 structures in Freestyle medium. 24 hours before transfection, 300 mL of 293E cells at  $0.5 \times 10^6$  cells/mL were seeded in a 1 L cell culture flask, and cultured in a 37° C., 5% CO<sub>2</sub> incubator with shaking at 120 rpm. During transfection, 300 µL of 293fectin™ was added to 5.7 mL Opti-MEM™. After mixing well, the mixture was incubated at room temperature for 2 minutes. Meanwhile, 300 µg of the expression plasmids for B2 and FV1 were diluted to 6 mL with Opti-MEM™, respectively. The diluted transfection reagent 293fectin and plasmids were mixed thoroughly and incubated at room temperature for 15 minutes. After that, the mixture was added to cells and mixed well, and cultured in a 37° C., 5% CO<sub>2</sub> incubator with shaking at 120 rpm for 7 days.

#### Example 2 Purification and Detection of Preferred Antibodies

##### (53) Purification of Proteins with B2 Structure

(54) The cell culture medium was centrifuged at 2000 g for 20 min, and the supernatant was collected and then filtered with a 0.22 micron filter membrane. Next, the supernatant was subjected to Protein L (GE) chromatography, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. After protein L chromatography, the sample was then subjected to affinity chromatography coupled with human PDL1 protein, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. Purified samples were separated by SDS-PAGE using 4-20% gradient gel (GenScript Biotechnology Co., Ltd.) to detect purified proteins. The results are shown in FIG. 5 and the purity of the preferred antibody B2 was 95%.

##### (55) Purification of Proteins with FV1 Structure

(56) The cell culture medium was centrifuged at 2000 g for 20 min, and the supernatant was collected and then filtered with a 0.22 micron filter membrane. Next, the supernatant was subjected to Mabselect Sure (GE) chromatography, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. After Mabselect Sure chromatography, the sample was then subjected to Protein L (GE) chromatography, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. After protein L chromatography, the sample was then subjected to affinity chromatography coupled with human CD47 protein, the proteins were eluted with 20 mM citrate-sodium citrate, pH 3.0, and then the resultant was adjusted to neutral pH with 1 M Tris base. Purified samples were separated by SDS-PAGE using 4-20% gradient gel (GenScript Biotechnology Co., Ltd.) to detect purified proteins. The results are shown in FIG. 5 and the purity of the preferred FV1 was 90.8%.

#### Example 3 ELISA Detection of Preferred Antibodies Binding to Human CD47 and Human PD-L1

(57) 1. Coating the first antigen: Human PD-L1-mFc (constructed by GeneScience, SEQ ID NO:



- (22) was diluted with PBS to 1  $\mu\text{g/mL}$ , and then added to a 96-well microtiter plate at 50  $\mu\text{L}$  per well and incubated overnight at 4° C.
- (58) 2. Blocking: After being washed three times, the plate was blocked with 3% BSA at 250  $\mu\text{L}$  per well, and incubated at 37° C. for 2 hours.
- (59) 3. Adding candidate antibody: After washing three times, the candidate antibody was added to the plate, each with 12 samples diluted at a 2-fold concentration gradient with an initial concentration of 10  $\text{mg/mL}$ , positive control or negative control was added at 50  $\mu\text{L}$  per well, and incubated at 25° C. for 1 hour.
- (60) 4. Adding the second antigen: Human CD47-His (constructed by GeneScience, SEQ ID NO: 23) was diluted with PBS to 10  $\mu\text{g/mL}$ , and then added to the 96-well microtiter plate at 50  $\mu\text{L}$  per well and incubated at 25° C. for 1 hour.
- (61) 5. Adding the secondary antibody: After washing three times, HRP-labeled streptavidin (1:10,000) was added to the plate at 50  $\mu\text{L}$  per well, and incubated at 25° C. for 1 hour.
- (62) 6. Color development: After being washed four times, TMB color development solution was added to the plate 50  $\mu\text{L}$  per well, and developed color shielded from light at room temperature for 10 minutes.
- (63) 7. Terminating: The stop solution was directly added to the plate at 50  $\mu\text{L}$  per well to terminate the reaction.
- (64) 8. Detection: After terminating the reaction, the microtiter plate was immediately put into the microplate reader. The OD value at 450 nm was measured, and the original data was saved for sorting. The results are shown in FIG. 6, showing that for the purified antibody B2,  $\text{EC}_{50}=0.5636$ ; for FV1,  $\text{EC}_{50}=1.662$ .

#### Example 4 Affinity Determination of Preferred Antibodies

(65) The affinities of antibodies with B2 structure and with FV1 structure were detected by Biacore T200 instrument. The specific protocols were as follows. Human PD-L1-His and human CD47-His were coupled to CM5 biosensor chip (GE), and then the antibodies of different concentrations were flowed through the chip at a flow rate of 30  $\mu\text{L/min}$ . The binding between the candidate antibody and antigen was performed with a binding time of 120 s and a dissociation time of 300 s. The kinetic fitting was performed using BIAevaluation software (GE), and the results of affinity constants were obtained as shown in Table 2 and Table 3. The affinities of B2 and FV1 with PD-L1 were 1.09E-10 M and 2.71E-10 M, respectively; and the affinities of B2 and FV1 with CD47 were 2.58E-8 M and 1.50E-8 M, respectively.

(66) TABLE-US-00002 TABLE 2 Results of affinity determination of candidate antibodies with PD-L1

| Antibody | K <sub>a</sub> (1/Ms) | K <sub>d</sub> (1/s) | KD (M)   | R <sub>max</sub> (RU) |
|----------|-----------------------|----------------------|----------|-----------------------|
| B2       | 3.95E+05              | 4.28E-05             | 1.09E-10 | 7.067                 |
| FV1      | 4.40E+05              | 1.19E-04             | 2.71E-10 | 28.06                 |

PD-L1 positive 9.08E+05 2.36E-04 2.60E-10 80 monoclonal antibody

(67) TABLE-US-00003 TABLE 3 Results of affinity determination of candidate antibodies with CD47

| Antibody | KD (M)   | R <sub>max</sub> (RU) |
|----------|----------|-----------------------|
| B2       | 2.58E-08 | 50.00                 |
| FV1      | 1.50E-08 | 60.00                 |

CD47 positive 2.36E-08 50.92 monoclonal antibody

(68) The foregoing is preferred embodiments of the present disclosure, however, it should be noted that some improvements and modifications can be made thereto by those ordinary skilled in the art without departing from the principles of the present disclosure, and these improvements and modifications should also be deemed to be within the protection scope of the present disclosure.

## Claims

1. A bivalent bispecific antibody, comprising a1) scFv2 that is a single-chain variable fragment (scFv) derived from an antibody that specifically binds to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1 in an order of CH1-partial hinge-flexible peptide scFv2, and b1) scFv1 that is a single-chain variable fragment (scFv) derived from

an antibody that specifically binds to a second antigen and a light chain constant region CL in an order of scFv1-CL; or a2) scFv2 that is a single-chain variable fragment specifically binding to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1 in an order of scFv2-flexible peptide-CH1-partial hinge, and b2) scFv1 that is a single-chain variable fragment specifically binding to a second antigen and a light chain constant region CL in an order of CL-scFv1; wherein the order is from N-terminal to C-terminal.

2. The bivalent bispecific antibody according to claim 1, wherein the flexible peptide comprises a sequence of (GGGGS (SEQ ID NO: 25).sub.n or (GGGGSAS (SEQ ID NO: 26)).sub.n, wherein n is an integer greater than 0, and the IgG1 partial hinge region linked to the flexible peptide comprises a sequence of EPKSCDK (SEQ ID NO: 24).

3. The bivalent bispecific antibody according to claim 1, wherein CL and CH1 form a heterodimer via a disulfide bond formed between the terminal cysteine residue in CL and the cysteine residue in the heavy chain hinge region.

4. The bivalent bispecific antibody according to claim 1, where in the bivalent bispecific antibody comprises two chains, 1) one chain has an amino acid sequence as shown in SEQ ID NO: 2, and the other chain has an amino acid sequence as shown in SEQ ID NO: 11; 2) one chain has an amino acid sequence as shown in SEQ ID NO: 1, and the other chain has an amino acid sequence as shown in SEQ ID NO: 10; 3) one chain has an amino acid sequence as shown in SEQ ID NO: 3, and the other chain has an amino acid sequence as shown in SEQ ID NO: 12; or 4) one chain has an amino acid sequence as shown in SEQ ID NO: 4, and the other chain has an amino acid sequence as shown in SEQ ID NO: 13.

5. A method for producing the bivalent bispecific antibody according to claim 1, comprising a) transforming a host cell with a first vector, comprising a nucleic acid encoding a scFv specifically binding to a first antigen, a flexible peptide, a heavy chain constant region CH1 and a partial hinge region of IgG1, and a second vector, comprising a nucleic acid encoding a scFv specifically binding to a second antigen and a light chain constant region CL; b) culturing the host cell under conditions that allow the synthesis of an antibody; and c) recovering the antibody from the culture.

6. A nucleic acid molecule encoding the bivalent bispecific antibody of claim 1.

7. A host cell, comprising a nucleic acid molecule encoding the bivalent bispecific antibody of claim 1.

8. A composition comprising the bivalent bispecific antibody of claim 1, wherein the composition is a pharmaceutical composition or a diagnostic composition.

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